StrawChoco

Problem Description

In the kingdom of StrawChoco, there are m Strawberries and n Chocolates. Each StrawChoco has a mood value that starts at 0.

StrawChocos often participate in the following three types of activities:

- A StrawChoco goes to the zoo with another StrawChoco.
- A StrawChoco goes shopping with another StrawChoco.
- A StrawChoco celebrates their birthday and invites others. The invitation strategy differs between Chocolates and Strawberries:
    1. A Chocolate invites only the last Strawberry they went to the zoo with. If they have never gone to the zoo, they invite no one.
    2. A Strawberry invites all StrawChocos they have ever gone shopping with.

As the king of StrawChoco kingdom, you must track the mood value of each StrawChoco, which changes during these events.

For Chocolates:

- Gain 5 mood points when visiting the zoo.
- Gain 1 mood point when shopping.
- Gain 1 mood point on their birthday, and an additional 5 if they invite someone.
- Gain 1 mood point when invited to a birthday.

For Strawberries:

- Gain 1 mood point when visiting the zoo.
- Gain 5 mood points when shopping.
- On their birthday, if they invite k StrawChocos, gain k mood points.
- Gain 1 mood point when invited to a birthday.
- Special rule: when visiting the zoo or shopping with a Chocolate, gain an additional 5 mood points.

The activity-related interfaces are already implemented in `event.h`. You are to implement the `Strawberry` and `Chocolate` classes in `strawchoco.h` and `strawchoco.cpp`, maintaining their mood values accordingly.

downloads.zip

Input Format

The first line contains three integers n, m, and l, representing the number of Chocolates, Strawberries, and the number of events.

All StrawChocos are numbered from 0 to $n + m - 1$. Chocolates are numbered from 0 to $n - 1$, and Strawberries from n to $n + m - 1$.

The next l lines each describe an event:

- $op = 0$, followed by two integers x and y: StrawChoco x and StrawChoco y go to the zoo.
- $op = 1$, followed by two integers x and y: StrawChoco x and StrawChoco y go shopping.
- $op = 2$, followed by one integer x: StrawChoco x celebrates their birthday.

Events occur in the order given.

Output Format

Output $n + m$ lines. The i-th line contains an integer representing the final mood value of the StrawChoco with ID $i - 1$.

Sample Input

```
2 2 7
0 0 1
1 0 2
0 1 3
2 0
1 2 3
1 2 0
2 2
```

Sample Output

```
14
11
27
12
```

Explanation:

- Chocolate 0: Went to the zoo once, shopping twice, had one birthday and invited Chocolate 1, was invited to Strawberry 2's birthday. Total mood = 14.
- Chocolate 1: Went to the zoo twice, invited to Chocolate 0's birthday. Total mood = 11.
- Strawberry 2: Went shopping three times (twice with Chocolate 0), had a birthday inviting Strawberry 0 and 3. Total mood = 27.
- Strawberry 3: Went to the zoo with Chocolate 1, went shopping once, invited to Strawberry 2's birthday. Total mood = 12.

Test Data & Constraints

There are 10 test cases, each worth 10 points.

Constraints for different cases:

- Cases 0–3: Only Chocolates, no birthdays.

- Cases 4–6: No birthday events.
- Cases 7–9: No special constraints.

For all test cases: $n, m, l \leq 1000$

Submission Format

Package `strawchoco.h` and `strawchoco.cpp` into a single **zip** file for submission. Other files will be ignored. **Ensure your files are in the root directory of the zip, not inside a subdirectory. In other words, extracting the zip should directly yield `.cpp`, `.h` files, etc., rather than a folder containing them.** The OJ system will place the provided files into your directory during compilation and execution.